

Question: 1

Research two actual network switches available in India (for example, from Cisco, TP-Link, or D-Link) and list their key specifications such as number of ports, speed, and price in a table?

Answer: Network Switches Specification Comparison

Specification Metric	Option 1: Cisco Business CBS250-24T-4G	Option 2: TP-Link TL-SG1024
Switch Type	Smart Managed / Layer 2+	Plug-and-Play Unmanaged
Total Number of Ports	28 Ports (24x Gigabit RJ-45 + 4x 1G SFP Uplinks)	24 Ports (24x Gigabit RJ-45 Ports)
Port Data Transfer Speed	10/100/1000 Mbps (Gigabit Ethernet)	10/100/1000 Mbps (Gigabit Ethernet)
Switching Capacity	56 Gbps	48 Gbps
Management Interface	Graphical Web UI, CLI, SNMP, and Mobile App	None (Self-negotiating plug-and-play installation)
Core Software Features	VLAN support (up to 4094), Layer 3 Static Routing, Access Control Lists (ACLs), DoS protection	IEEE 802.3x Flow Control, Auto MDI/MDIX crossover detection
Form Factor & Cooling	Rack-mountable (1U size) & Fanless Design (Silent operation)	Rack-mountable / Desktop & Fanless Design (Silent operation)

Estimated

Price Range ₹25,000 - ₹31,000 INR
(India)

₹6,000 - ₹8,000 INR

Question: 2

Create a device selection matrix comparing at least one switch, one router, and one firewall that could be used for a small startup office (20-30 employees). Include columns for brand/model, key specs, price, and your rating for each device.
Hint: Use online sources like Amazon India, Flipkart, or manufacturer websites for real-world pricing and specs?

Answer: Small Startup Office (20-30 Employees) Device Matrix

Device Type	Brand & Model	Key Specifications (Specs)	Estimated Price Range (India)
Network Switch	TP-Link TL-SG1024DE	• 24x Gigabit RJ45 Ports • Easy Smart Managed Layer 2 • VLAN Support & Traffic Prioritization • Fanless Silent Steel Case	₹6,999 - ₹7,400 INR
Business Router	Cisco RV340 Dual WAN	• Dual-WAN Ports for Load Balancing • 4x Gigabit LAN Ports + 2x USB (3G/4G)	₹16,500 - ₹23,500 INR

		• 900 Mbps IPsec VPN Throughput • High-speed routing with content filtering	
		• 8x 1GbE Ports Security Gateway • 2 Gbps Firewall Throughput • 750 Mbps Advanced Threat Prevention • TLS 1.3 Encryption & Cloud Sandbox	₹28,500 - ₹38,000 INR (Hardware Only)
Hardware Firewall	SonicWall TZ270 Gen 7		

Question: 3

Imagine you are setting up the network for a new gaming café with 25 computers and public Wi-Fi. Select one switch, one router, and one firewall model for this scenario and justify your choices in 2-3 lines each?

Answer: 1. Switch: Cisco CBS350-48T-4G — It provides enough Gigabit Ethernet ports for 25 computers and allows extra ports for future devices. It gives fast and reliable LAN connectivity for gaming.

2. Router: TP-Link Archer AX73 — It provides high-speed Wi-Fi for customers and supports many devices. It is suitable for both gaming and public Wi-Fi.

3. Firewall: Fortinet FortiGate 40F — It protects the gaming café network from unauthorized access, malware, and other online threats while allowing secure internet access.

Question: 4

Prepare a Bill of Materials (BOM) listing the selected network devices (switch, router, firewall) for your gaming café scenario, including quantity, model, unit price, total price, and a short justification for each item?

Answer: Switch: Cisco CBS350-48T-4G

Quantity: 1

Unit Price: ₹45,000

Total: ₹45,000

Justification: Provides enough Gigabit ports for 25 gaming computers.

Router: TP-Link Archer AX73

Quantity: 1

Unit Price: ₹12,000

Total: ₹12,000

Justification: Provides fast Wi-Fi for customers and multiple devices.

Firewall: Fortinet FortiGate 40F

Quantity: 1

Unit Price: ₹55,000

Total: ₹55,000

Justification: Protects the network from unauthorized access and cyber threats.

Grand Total: ₹1,12,000

Question: 5

Perform a cost-benefit analysis for your chosen devices: explain in 3-4 lines why you selected these models over cheaper or more expensive alternatives, considering both features and budget?

Answer: 1. Switch - Cisco CBS350-48T-4G

It is more expensive than basic switches but provides many Gigabit ports.

It is reliable and suitable for 25 gaming PCs.

The higher cost is justified by better performance and future expansion.

2. Router - TP-Link Archer AX73

It offers high-speed Wi-Fi and supports many devices.

It is reasonably priced compared with high-end gaming routers.

It provides good performance while keeping the café budget under control.

3. Firewall - Fortinet FortiGate 40F

It costs more than basic security solutions but provides strong network protection.

It helps protect the café from cyber threats and unauthorized access.

The extra cost is worthwhile because security is important for a public gaming cafe.